7.2 Hydrocarbons (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.2 Hydrocarbons (A Level Only)
Difficulty	Hard

Time allowed: 50

Score: /36

Percentage: /100

Question 1a

The nitration of benzene is the first important step in manufacturing dyes and explosives.

i)
Write the equation for the generation of the electrophile.

[1]

ii)
State which reactant acting as a Brønsted-Lowry base and justify your answer.

[1]

iii)
Identify the conjugate acid in the reaction.

[1]

[3 marks]

Question 1b

Compound **B** is produced in two steps as outlined in Fig. 1.1.

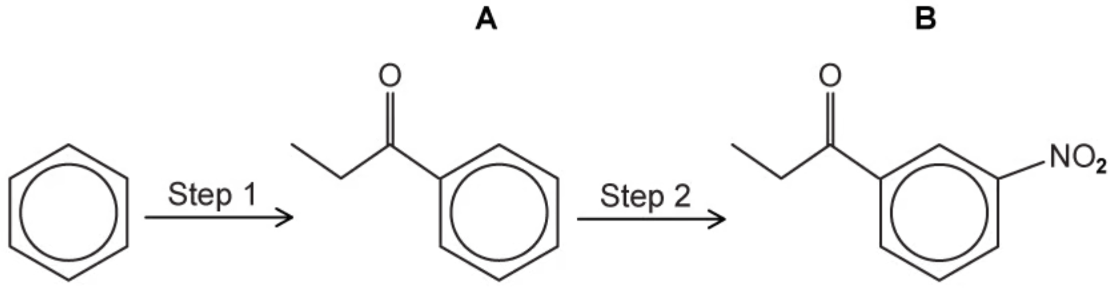


Fig. 1.1

i)
Outline the reagents and conditions required for the production of compound **A** drawn in Fig. 1.1.

[2]

ii)
Using curly arrows, describe the mechanism for step **1**.

[3]

[5 marks]

Question 1c

Draw the dot-and-cross diagram for the structure of the catalyst, once the electrophile has been generated in part **(b)**.

[2 marks]

Question 1d

Explain why benzene can generally only undergo substitution reactions.

[4 marks]

Question 2a

Fig. 2.1 shows the formation of a ketone followed by an alcohol.

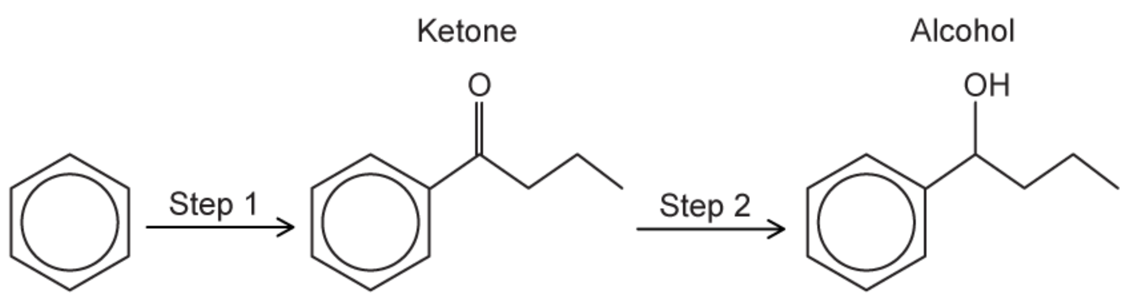


Fig. 2.1

Draw the mechanism to show the formation of the catalyst responsible for step **1**.

[2 marks]

Question 2b

Draw the mechanism for the formation of the ketone, including the regeneration of the catalyst required.

[5 marks]

Question 2c

State the reagents required for step **2**.

[2 marks]

Question 3a

Benzene can be converted into cyclohexane as shown in Fig. 3.1.

Fig. 3.1

i)
For this reaction, name the type of reaction and identify the reagent and conditions needed.

type of reaction ....

reagent and conditions ....

[2]

ii)
State the bond angles in benzene and cyclohexane.

bond angle in benzene ....

bond angle in cyclohexane ....

Explain your answers.

[2]

[4 marks]

Question 3b

When benzene reacts with SO_3, as shown in Fig. 3.2, benzenesulfonic acid is produced.

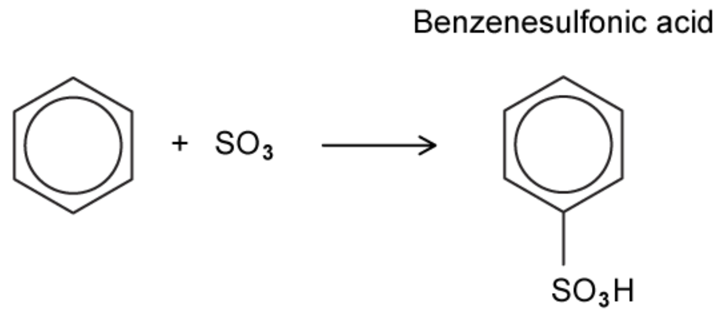


Fig. 3.2

The mechanism of this reaction is similar to that of the nitration of benzene. Concentrated H_2SO_4 is used in an initial step to generate the SO_3H^+ electrophile as shown.

$$SO_3 + H_2SO_4 \rightarrow SO_3H^+ + HSO_4^-$$

i)

Draw a mechanism for the reaction of benzene with SO_3H^+ ions in Fig. 3.3. Include all necessary curly arrows and charges.

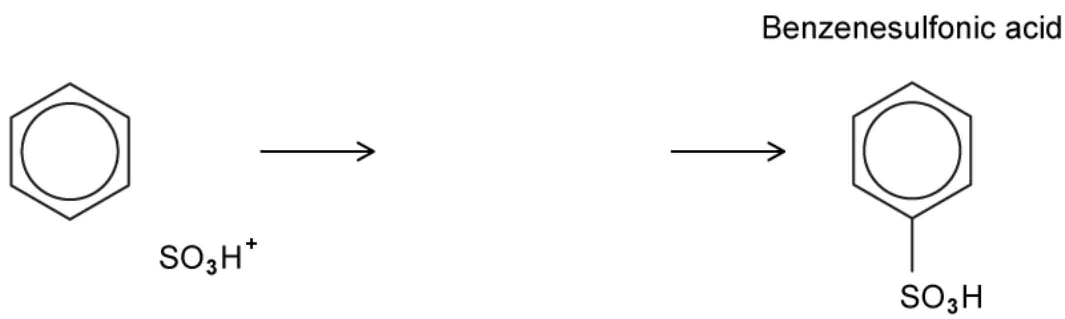


Fig. 3.3

[3]

ii)

Write an equation to show how the H_2SO_4 catalyst is reformed.

[1]

[4 marks]

Question 3c

3–aminobenzoic acid can be synthesised from methylbenzene in three steps as shown in Fig. 3.4.

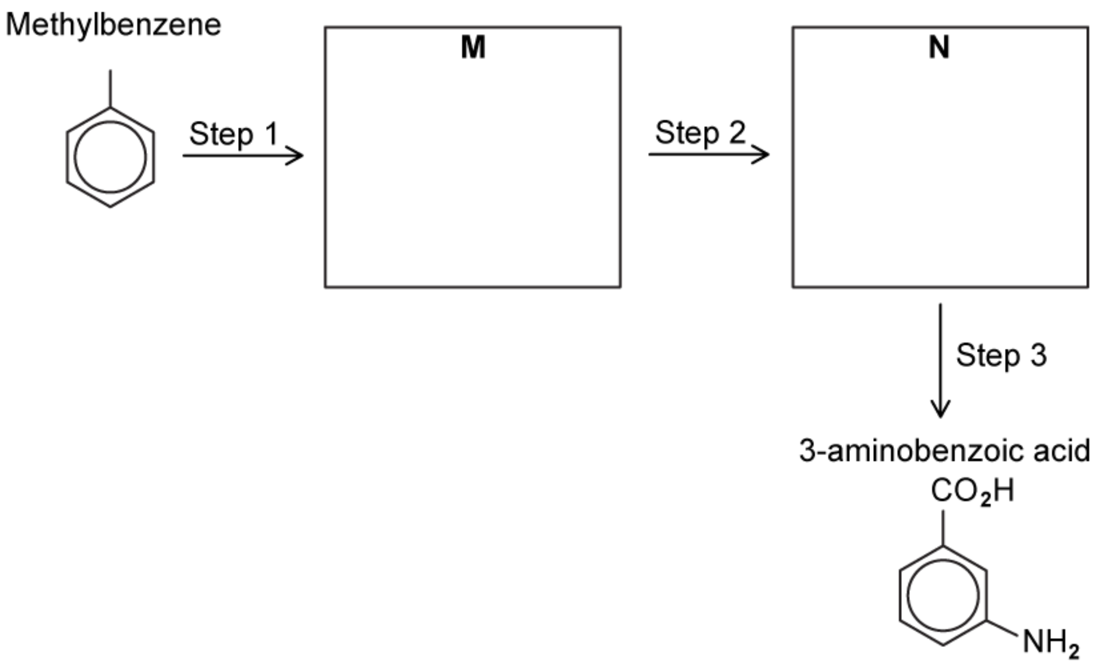


Fig. 3.4

i)
Draw the structures of **M** and **N** in the boxes.

[2]

ii)
Suggest reagents and conditions for each step of the synthesis.

step **1** ....

step **2** ....

step **3** ....

[3]

[5 marks]

Model Answers: Hard

1a

a)

i) The generation of the nucleophile for the nitration of benzene is:

- $2H_2SO_4 + HNO_3 \rightarrow 2HSO_4^- + H_3O^+ + NO_2^+$; [1 mark]

ii) The reactant that is acting as a Brønsted-Lowry base is:

- Nitric acid / HNO_3
  AND
  It accepts a proton from sulfuric acid / H_2SO_4; [1 mark]

iii) The conjugate acid is:

- H_3O^+
  AND
  NO_2^+; [1 mark]

[Total: 3 marks]

- **Remember:** You can have more than one species in a conjugate acid or base pair
- The overall equation for the reaction is:
  - $2H_2SO_4 + HNO_3 \rightarrow 2HSO_4^- + H_3O^+ + NO_2^+$
- H_2SO_4 donates a proton to become HSO_4^-
  - This means it acts as a Brønsted-Lowry acid
  - Therefore, its conjugate base is HSO_4^-
- HNO_3 accepts a proton to become $H_2NO_3^+$ which will exist as H_3O^+ and NO_2^+
  - This means that the HNO_3 acts as a Brønsted-Lowry base
  - Therefore, H_3O^+ and NO_2^+ are the conjugate acids

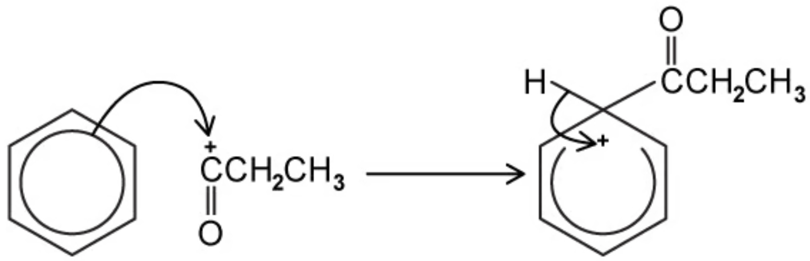
1b

b)

i) The reagents and conditions required for step **1** are:

- Aluminium chloride / $AlCl_3$; [1 mark]
- Propanoyl chloride / CH_3CH_2COCl; [1 mark]

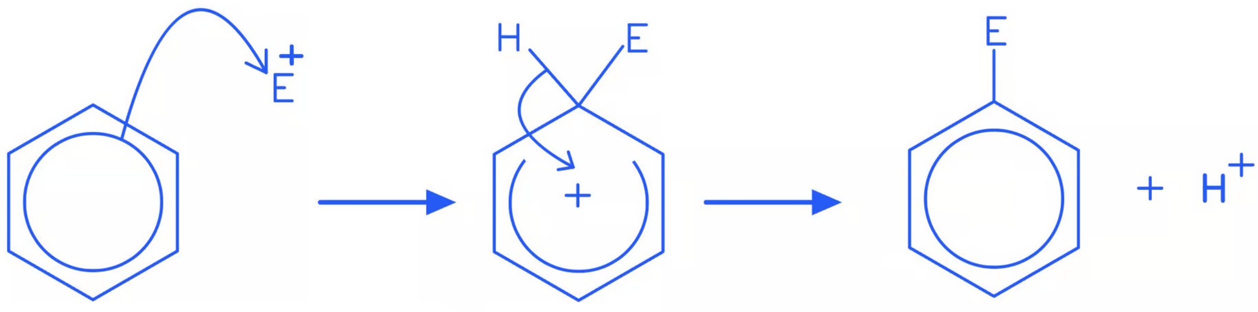
ii) The mechanism for step **1** is:



- A curly arrow from the ring inside benzene to the C+; [1 mark]
- Correct intermediate structure with horseshoe covering more than half of the benzene but not extending past carbon 2 or 6; [1 mark]
- A curly arrow from the C-H bond back to the + within the ring; [1 mark]

[Total: 5 marks]

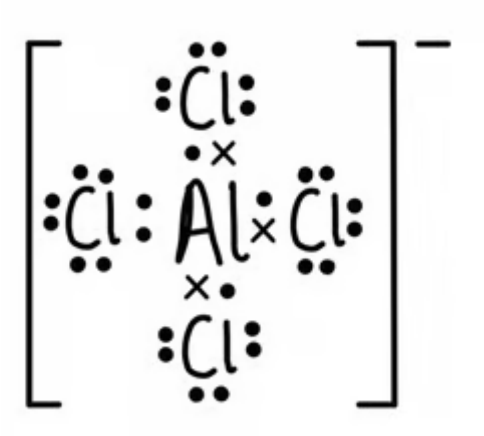
- There are extra marking points for electrophilic substitution mechanisms:
  - The + must be on the correct carbon in the $COCH_3CH_2$ electrophile
  - The first arrow must start touching the ring, not start inside the ring
  - The horseshoe in the ring must be centred about carbon 1 and not extend past carbon 2 or 6
- All electrophilic substitution mechanisms follow the same general pattern:



- The electrophile bonds to the benzene ring, forming an intermediate with a partially delocalised electron system - shown as a horseshoe with + inside
- The intermediate loses a proton, restoring the delocalised electron system

1c

c) The dot-and-cross diagram for the structure of the catalyst, once the electrophile has been generated is:



- Correct bonds between Al-Cl atoms; [1 mark]
- Correct number of electron pairs around Cl atoms; [1 mark]

[Total: 2 marks]

- The equation for the generation of the ion for step **1** is:
  - $CH_3CH_2COCl + AlCl_3 \rightarrow CH_3CH_2CO^+ + AlCl_4^-$
- In $AlCl_3$, the aluminium is electron deficient because it has six electrons in the outer shell
- The chlorine atom leaving the acyl chloride bonds to the $AlCl_3$
  - This forms $AlCl_4^-$ and generates the $CH_3CH_2CO^+$ electrophile

1d

d) Benzene can only undergo substitution reactions because:

- Addition reactions require delocalised electrons to form new bonds; [1 mark]
- Which requires energy; [1 mark]
- The C-H bond is not involved in delocalisation in the benzene ring; [1 mark]
- Which can remain if a hydrogen atom is substituted; [1 mark]

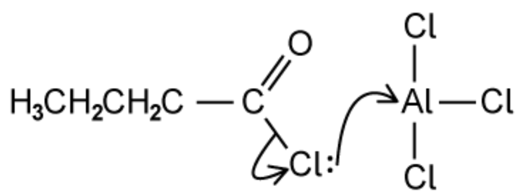
[Total: 4 marks]

- Because of the delocalised electrons exposed above and below the plane of

the rest of the molecule, benzene is going to be highly attractive to electrophiles
  - **Remember:** Electrophiles are species which seek electron-rich areas in other molecules
- The benzene ring can maintain the delocalisation if it replaces a hydrogen atom with another species - a substitution reaction
- The hydrogen atoms aren't involved with the delocalised electrons

2a

a) The mechanism for the formation of the catalyst in step **1** is:



- Curly arrow from the C-Cl bond to the Cl; [1 mark]
- Curly arrow from one of the Cl lone pairs to the $AlCl_3$; [1 mark]

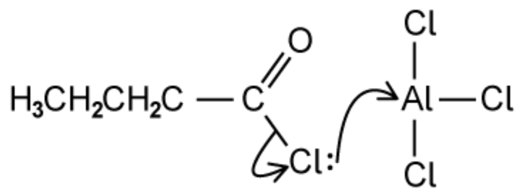
[Total: 2 marks]

- For this reaction, the generation of a $CH_3CH_2CH_2CO^+$ electrophile is required
- In order to do this, a catalyst is required
- The equation for the generation of the electrophile is:

- The hydrogen atoms aren't involved with the delocalised electrons

2a

a) The mechanism for the formation of the catalyst in step **1** is:



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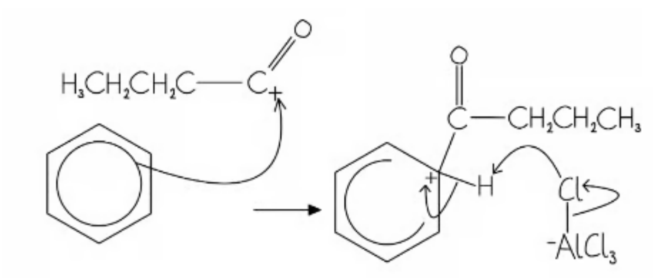
[Total: 2 marks]

- For this reaction, the generation of a $CH_3CH_2CH_2CO^+$ electrophile is required
- In order to do this, a catalyst is required
- The equation for the generation of the electrophile is:
  - $CH_3CH_2CH_2COCl + AlCl_3 \rightarrow CH_3CH_2CH_2CO^+ + AlCl_4^-$
- If you know the equation, then you can use this to work out the mechanism
- From this equation:
  - You know that the chlorine is leaving the butanoyl chloride, so that bond must break
  - You know that $AlCl_4^-$ is forming, so move the chlorine over using a curly arrow

2b

b) The mechanism for the formation of the ketone is:



- Curly arrow from the ring inside benzene to the C+ of the carbocation; [1 mark]
- Correct intermediate structure; [1 mark]
- Curly arrow from the C-H bond back to the + within the ring; [1 mark]

- Curly arrow from Al-Cl bond to Cl atom; [1 mark]
- Curly arrow from Cl atom to H atom in the C-H bond to regenerate the nucleophile; [1 mark]

[Total: 5 marks]

- There are extra marking points for electrophilic substitution mechanisms:
  - The + must be on carbon-1 of the electrophile
  - The first arrow must start touching the ring, not start inside the ring
  - The horseshoe must cover at least half of the ring with the 'gap' at the correct carbon, i.e. it must be centred about carbon 1 and not extend past carbon 2 or 6
- The final products will be HCl, $AlCl_3$ (the regenerated catalyst) and the ketone

2c

c) The reagents required for step **2** are:

- Sodium borohydride / sodium tetrahydridoborate / $NaBH_4$; [1 mark]
- Ethanol / CH_3CH_2OH; [1 mark]

[Total: 2 marks]

- In an aqueous solution, $NaBH_4$ generates the hydride ion nucleophile, $:H^-$
- The hydride ion will reduce a carbonyl group in a ketone, but is not strong enough to reduce a C=C double bond
  - This is because it is attracted to the C in the C=O bond, but is repelled by the high electron density of the C=C bond
- This requires an ethanol solvent

3a

a)

i) The type of reaction, reagents and conditions are:

- Type of reaction = reduction / hydrogenation / addition; [1 mark]
- Reagents and conditions = H_2
  AND
  Ni / Pt catalyst; [1 mark]

ii) The bond angles are:

- Benzene = 120°
  AND
  Cyclohexane = 109.5°; [1 mark]
- As π-bonds are transformed into σ-bonds; [1 mark]

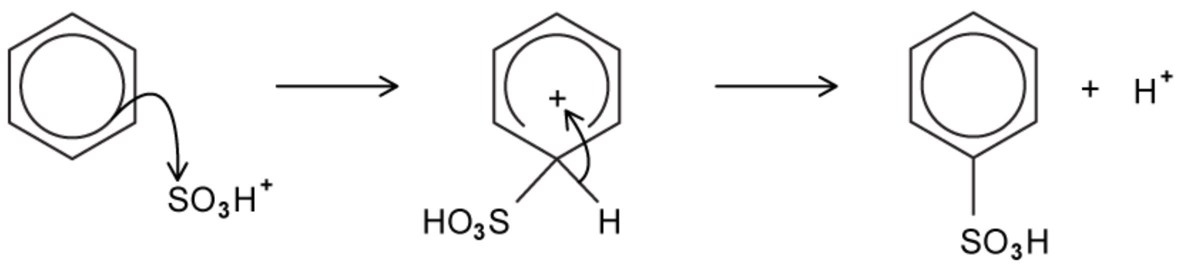
[Total: 4 marks]

- The hydrogenation of benzene is a reduction reaction
- Benzene is heated with hydrogen gas and a nickel or platinum catalyst to form cyclohexane
  - This is a reaction that you need to learn and remember that reducing reagents are different for different reactions
- Benzene and other aromatic compounds are regular and planar compounds with bond angles of 120°
- In benzene, each carbon atom in the ring forms three σ bonds using the sp^2 orbitals and the remaining p orbital overlaps laterally with p orbitals of neighbouring carbon atoms to form a π bond
- In cyclohexane, there are 4 σ bonds, so the bond angle must now be 109.5°

3b

b)

i) The mechanism is:



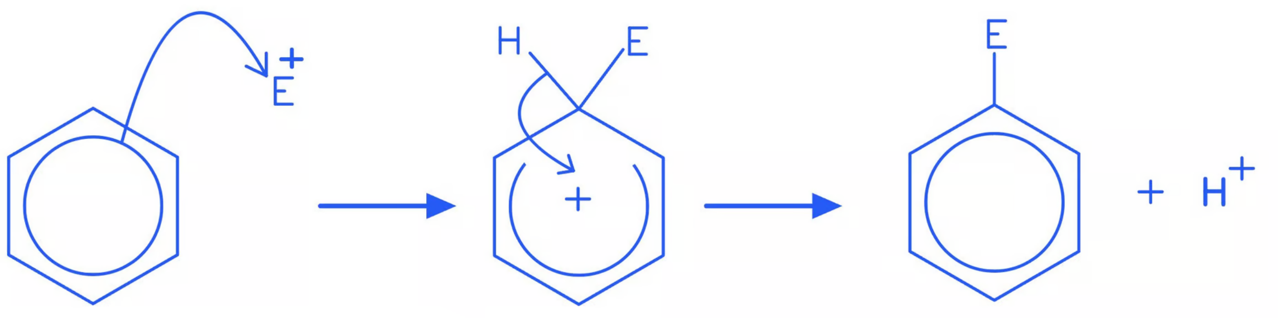
- Curly arrow to the sulfur atom; [1 mark]
- Correct intermediate shown; [1 mark]
- Curly arrow from the C-H bond to the + inside the ring
  AND
  H^+ formed / lost; [1 mark]

ii) The equation to reform the catalyst is:

- $HSO_4^- + H^+ \rightarrow H_2SO_4$; [1 mark]

[Total: 4 marks]

- This mechanism is not one that you need to specifically know
  - However, you should be aware that the electrophilic substitution of a hydrogen atom on a benzene ring has the same mechanism regardless of the electrophile
- **Careful:** Make sure the atom with the positive charge is bonded to the carbon atom on the benzene ring
  - For this question, this should be the sulfur atom



- The question tells you that H_2SO_4 is used to generate the electrophile
  - This means it is a catalyst, similar to how a halogen carrier works in the alkylation and acylation of a benzene ring

- $HSO_4^- + H^+ \rightarrow H_2SO_4$; [1 mark]

[Total: 4 marks]

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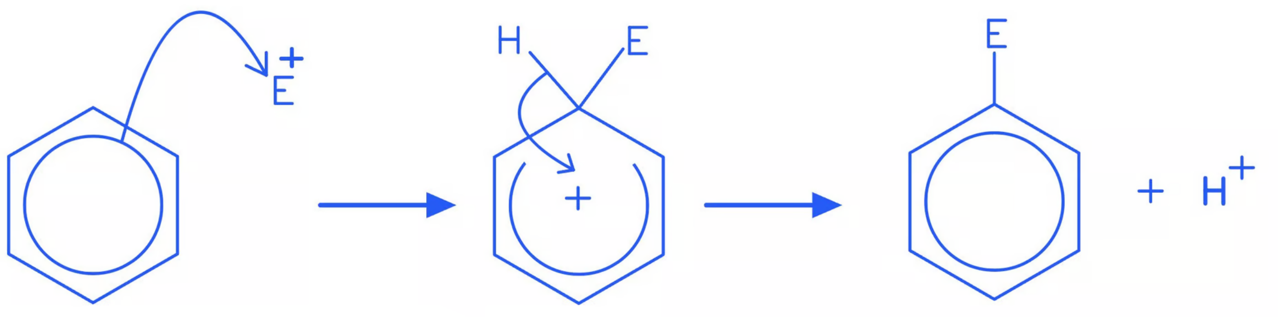


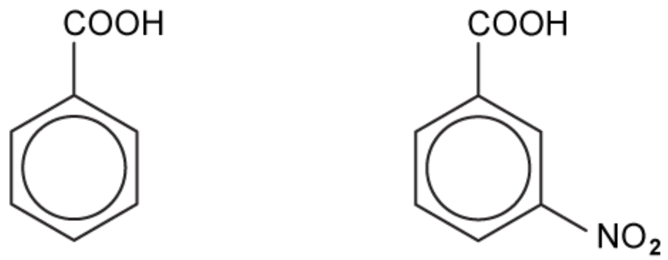
- The question tells you that H_2SO_4 is used to generate the electrophile
  - This means it is a catalyst, similar to how a halogen carrier works in the alkylation and acylation of a benzene ring
- The H^+ ion which is generated from the reaction reacts with HSO_4^- to reform the concentrated sulfuric acid, H_2SO_4

3c

c)

i) The structures of **M** and **N** are:

M **N**



- Correct structure of **M**: [1 mark]

- Correct structure of **N**; [1 mark]

ii) The reagents and conditions for steps 1, 2 and 3 are:

- Step **1** = hot $KMnO_4$ / MnO_4^-; [1 mark]
- Step **2** = conc. H_2SO_4
  AND
  Conc. HNO_3; [1 mark]
- Step **3** = Sn
  AND
  Conc. HCl (heat); [1 mark]

[Total: 2 marks]

- In step **1**, the alkyl side-chains in alkyl arenes are oxidised to carboxylic acids when refluxed with alkaline potassium manganate(VII) and then acidified with dilute sulfuric acid (H_2SO_4)
- The second step involves nitration of the benzene ring in the 3– position as the COOH group is electron withdrawing
- In the final step, the NO_2 group is reduced using Sn and concentrated HCl
  - **Careful:** Reducing agents are not a 'one size fits all' reagent
    - Make sure you learn the specific reducing agents for their reactions